

Amylase Rate, Graph and Topic 1 Consolidation

LAUNCH • Science • 40 minutes

I can calculate amylase rate from endpoint time, plot and interpret the pH pattern, evaluate the method, and apply Topic 1 knowledge to exam-style evidence.

Prepare before pupils arrive

Setup: Prepare an intact raw dataset and an accessible backup dataset, graph templates and calculators. Decide whether pupils use $\text{rate} = 1/t$ or a clearly labelled scaled convention; never mix both in one table.

Safety: Do not infer CP completion from a graph or generated printout. Preserve pupil/raw evidence, avoid public ranking by practical outcome and treat anomalous data as a reasoning opportunity rather than misconduct.

Equipment: Pupil or teacher-provided raw dataset, calculator, graph paper or local spreadsheet, ruler, command-word frame and Topic 1 retrieval grid.

Safety: Analysis uses no reagents. Keep original raw data unchanged, use anonymised/class data only and follow school device/privacy rules. Do not present the analysis page as proof that the wet practical was completed.

Fallback: Use a pre-scaled graph, rate choice cards and an adult-scribed evaluation from the pupil's exact reasoning; mathematical and command-word goals remain visible.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation • Paper 1 • Topic 1 • rate calculation, CP 1.10 analysis and Topic 1 consolidation.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 3 min	Explain why shorter endpoint time means faster amylase activity. State one controlled variable from the investigation.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	reciprocal — one divided by optimum — highest measured activity evaluate — weigh evidence and judge Write the relationship before substituting: $\text{rate} = 1/\text{time}$. Keep endpoint time in seconds unless the question states otherwise.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Plot pH on the independent-variable x-axis and rate on the y-axis. Use an even scale, units and a sensible smooth trend only when the data justify it.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each data statement to calculation check, pattern/conclusion or evaluation before writing the full response. Choose one evidence statement.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Complete a GCSE-style CP analysis: rate table, pH-rate graph, evidence-led conclusion, method evaluation and a compact Topic 1 transfer check. Record: Shown rate working, an accurately labelled graph or equivalent data interpretation, a bounded conclusion, a strength-limitation-improvement chain and one Topic 1 transfer response.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Use one graph feature to explain how pH affected amylase, then name one justified method improvement. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: 0.00800 s^{-1}

Correct. $1 \div 125 = 0.00800 \text{ s}^{-1}$ to three significant figures.

Misconception repair

Repaired. The conclusion is bounded by tested pH values, repeats and measurement uncertainty.

Guided checkpoint

Correct. $1 \div 125 = 0.00800 \text{ s}^{-1}$ to three significant figures. Use rate = $1/\text{time}$, not rate = time and not $1000/\text{time}$ unless the question explicitly gives that scaled convention.

Reciprocal value

CALCULATION CHECK. Calculation: the equation, substitution, value and unit agree. This card checks a numerical rate, not the whole pH pattern.

Direction check

CALCULATION CHECK. Calculation check: this tests whether time and rate were inverted correctly. Use this statement to validate the numerical transformation.

Highest tested rate

PATTERN / CONCLUSION. Conclusion: pH 7 is the optimum among the conditions tested, not necessarily an exact universal value. This statement interprets the graph rather than judging the method.

Rise then fall

PATTERN / CONCLUSION. Pattern: the description covers both sides of the measured peak. This is what the plotted values show.

Subjective endpoint

EVALUATION. Evaluation: subjective colour judgement can reduce repeatability and precision. This card identifies a limitation of evidence collection.

Shorter interval

EVALUATION. Evaluation: the improvement targets a named timing limitation. This changes method precision, not the biological conclusion directly.

Preserve anomaly

EVALUATION. Evaluation: an anomalous value requires transparent checking, not silent deletion. This card concerns evidence quality and judgement.

Model limitation

A smooth curve can hide repeats, anomalies and uncertainty. The graph covers only tested pH values and does not by itself prove the molecular mechanism or an exact optimum between points.

Independent evidence

Shown rate working, an accurately labelled graph or equivalent data interpretation, a bounded conclusion, a strength-limitation-improvement chain and one Topic 1 transfer response.

Supported: Calculate three rates from a worked equation, plot supplied coordinates, choose the tested optimum and complete one limitation-improvement pair. Use a calculator sequence, partially labelled axes and stems: The fastest was ___ because ___. A limitation was ___, so improve by ___.

Standard: Calculate the full rate column, plot the graph, describe and explain the pattern, then evaluate the endpoint method and answer one Topic 1 exam question. Follow CALCULATE → DESCRIBE → EXPLAIN → EVALUATE; keep each command-word response distinct.

Stretch: Analyse repeats and uncertainty, justify the tested optimum, evaluate whether the conclusion is valid and transfer enzyme/transport knowledge to an unfamiliar context. Quantify where possible; distinguish accuracy, precision, repeatability and validity without claiming more than the evidence supports.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W8L3_Amylase_Rate_And_Topic_1_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.